# Supplementary Material: Order-Aware Temporal Dynamics in Disorders of Consciousness: Multi-Scale Electrophysiological Modeling and Multicenter Evaluation

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## SUPPLEMENTARY CLINICAL AND METHODOLOGICAL NOTES

### A. CRRC Cohort Recruitment and Ethics Compliance

The prospective inpatient cohort was evaluated at the China Rehabilitation Research Center (CRRC / Beijing Bo'ai Hospital). Inpatients presenting with disorders of consciousness secondary to severe acute brain injury (11 ICH, 4 TBI) were enrolled during neurorehabilitation. Ethical approval was granted by the CRRC Medical Ethics Committee (Approval No. 2025-118-01; see Duan et al. [1]). Written informed surrogate consent was obtained from legally authorized representatives in accordance with the Declaration of Helsinki. Diagnostic categorization into minimally conscious state (MCS/EMCS, $N = 6$; 5 MCS and 1 EMCS) and vegetative state / unresponsive wakefulness syndrome (VS/UWS, $N = 9$) was established through serial Coma Recovery Scale-Revised (CRS-R) assessments by certified clinicians (scores: 2–23). All recordings were conducted during stable post-acute rehabilitation without continuous acute sedation.

### B. Telemetry Acquisition and Signal Preprocessing

Bedside electroencephalography was acquired using a NeuSen NSM2 amplifier / EEGCap32-pony system (1,000 Hz, 19-channel 10–20 montage; Duan et al. [1]). Continuous recordings spanned intermediate intervals (mean $19.53 \pm 7.96$ min, range 7.27–39.07 min). Raw continuous signals were re-referenced to Common Average Reference (CAR) and segmented into non-overlapping 4-s blocks (4,000 samples per block).

Artifact screening operated at the individual recording level without data leakage: blocks with peak-to-peak amplitude $\geq 500\,\mu$V were flagged as artifactual gaps ($\mathbb{I}_{\text{adj}} = 0$). Homologous electrodes matching the discovery montage (F3, F4, C3, C4, O1, O2) were extracted. Two subjects (CRRC_008 and CRRC_009) exhibited persistent high-impedance drift on $O_2$. Channel interpolation (e.g., spherical splines) was deliberately avoided to preclude artificial bilateral covariances across sparse arrays; instead, two protocols evaluate sensor robustness: (1) **Intact 6-Channel Subset** ($N = 13$): excludes localized $O_2$ drift; (2) **Full Prospective Cohort** ($N = 15$): evaluates all 15 enrolled subjects under strict intent-to-treat.

### C. Matched Cross-Center Preprocessing, Direction Alignment, and Domain Adaptation

All matched comparisons evaluate identical feature matrices across models: $\mathbf{x} = [c_{\text{global}}, -\text{CV}_\theta, \rho_1, r_{\text{residual}}]^\top$. Source inner cross-validation selects ridge and logistic regularization; LDA uses analytical shrinkage. The source sign vector is $\mathbf{s}_{\text{source}} = [+1, +1, +1, -1]^\top$.

Three cross-center protocols are evaluated on CRRC telemetry: (1) **Source-Frozen Transfer**: retains source moments and signs (0.5741 AUROC); (2) **Target-Standardized Transfer**: standardizes features via target batch moments with source signs (0.5000 AUROC); (3) **Target-Aligned Transfer**: aligns feature directions for subacute rehabilitation ($\mathbf{s}_{\text{target}} = [-1, +1, +1, +1]^\top$), achieving 0.7778 AUROC (0.7963 with $c = 1.5$ truncation; Table S2). Biophysically, this sign inversion reflects pathophysiological divergence: acute coma exhibits burst suppression in VS/UWS and emerging theta pacing in MCS ($s_1 = +1$). In subacute rehabilitation (¿1 month post-injury), chronic VS/UWS maintains diffuse slow waves while recovering MCS restores posterior alpha rhythms, lowering $c_{\text{global}}$ ($s_1 = -1$). Concurrently, acute VS/UWS displays diffuse slow-wave hyper-synchrony ($s_4 = -1$), whereas subacute focal lesions (11/15 CRRC patients) isolate hemispheres via callosal disruption, leaving residual correlation higher in conscious MCS ($s_4 = +1$). For prospective unlabeled deployment, directions align via the *unsupervised anchor rule*: $s_k^{\text{target}} = \text{sign}(\widehat{\text{Corr}}_T(x_k, \rho_1))$, anchored to autoregressive persistence ($\rho_1$). Source-training medians yield 0.6667 sensitivity and 0.7778 specificity on CRRC; Figshare source-frozen screening detects 19/19 MCS cases (sensitivity 1.0000, specificity 0.1290; Youden-calibrated: sensitivity 0.7368, specificity 0.7097). Transforms and hashes are archived in `results/matched_model_comparison_v1/`; paired uncertainty uses 10,000 Dirichlet draws.

### D. Bedside Streaming Implementation, Computational Benchmarking, and Latency Deconstruction

To evaluate translational feasibility, the four-feature consensus pipeline was benchmarked in a streaming architecture prototype (`docs/bedside_cdss_demo.html`; animation: `docs/bedside_playback_demo.gif`). The architecture ingests non-overlapping 4-s epochs ($f_s = 256\,$Hz) with automated artifact gating ($< 500\,\mu$V). Physical adjacency

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TABLE S1

DE-IDENTIFIED INDIVIDUAL COHORT CHARACTERISTICS, FEATURE PROFILES, AND CROSS-CENTER BENCHMARK PERFORMANCE FOR THE CHINA REHABILITATION RESEARCH CENTER (CRRC / BEIJING BO'AI HOSPITAL) INPATIENT COHORT ($N = 15$; 15 MALES; 6 MCS [INCLUDING 1 EMCS], 9 VS/UWS; 11 ICH, 4 TBI; MEAN AGE $51.7 \pm 12.3$ YEARS). CLEAN 6-CHANNEL RECORDS ($N = 13$; 6 MCS [INCLUDING 1 EMCS], 7 VS/UWS) EXCLUDE CRRC_008 AND CRRC_009 DUE TO CONTINUOUS MECHANICAL ELECTRODE DRIFT ON $O_2$.

| Subject ID | Diagnosis | CRS-R | Lesion Laterality | Duration (s) | Retention | $c_{\text{global}}$ | $-\text{CV}_\theta$ | $\rho_1$ | $r_{\text{residual}}$ |
|---|---|---|---|---|---|---|---|---|---|
| CRRC_001 | MCS | 11 | Bilateral intact | 605.0 | 98.7% | 0.1834 | -1.3927 | +0.1118 | +0.5878 |
| CRRC_002* | MCS$^{+}$* (EMCS) | 23 | Right hemisphere | 1236.0 | 98.7% | 0.5343 | -0.3331 | +0.3201 | +0.2064 |
| CRRC_003 | MCS | 10 | Left hemisphere | 1225.0 | 97.1% | 0.8409 | -0.3682 | +0.1451 | +0.3646 |
| CRRC_004 | MCS | 7 | Left hemisphere | 1221.0 | 100.0% | 0.5993 | -0.3748 | -0.0378 | +0.1391 |
| CRRC_005 | MCS | 17 | Bilateral intact | 845.0 | 100.0% | 0.4906 | -0.6205 | +0.3850 | +0.3328 |
| CRRC_006 | MCS | 9 | Right hemisphere | 1216.0 | 97.4% | 0.5173 | -0.4568 | +0.5073 | +0.2188 |
| CRRC_007 | VS/UWS | 7 | Right hemisphere | 1896.0 | 99.8% | 0.8119 | -0.4731 | +0.0105 | +0.3776 |
| CRRC_008$^{\dagger}$ | VS/UWS | 2 | Bilateral intact | 1213.0 | 100.0% | 0.7253 | -0.6384 | +0.0337 | +0.3048 |
| CRRC_009$^{\dagger}$ | VS/UWS | 3 | Bilateral intact | 1209.0 | 99.7% | 0.6213 | -0.6091 | +0.0614 | +0.1660 |
| CRRC_010$^{\ddagger}$ | VS/UWS | 2 | Bilateral intact | 637.0 | 21.4% | 0.7294 | -0.4525 | +0.2401 | -0.0657 |
| CRRC_011 | VS/UWS | 7 | Bilateral intact | 1237.0 | 99.0% | 0.4154 | -0.4882 | +0.4003 | +0.3513 |
| CRRC_012 | VS/UWS | 7 | Bilateral intact | 436.0 | 99.1% | 0.5607 | -0.3266 | +0.0680 | +0.0339 |
| CRRC_013 | VS/UWS | 7 | Left hemisphere | 2344.0 | 99.5% | 0.5517 | -0.4399 | +0.1056 | +0.2702 |
| CRRC_014 | VS/UWS | 7 | Right hemisphere | 1215.0 | 86.8% | 0.7556 | -0.6032 | +0.6941 | +0.1504 |
| CRRC_015 | VS/UWS | 4 | Left hemisphere | 1040.0 | 100.0% | 0.6218 | -0.7362 | -0.0033 | +0.3001 |

*CRRC_002 met Emergence from MCS (EMCS) criteria; retained in conscious class (MCS$^{+}$) for binary triage. $^{\dagger}$Mechanical $O_2$ electrode drift; excluded in Protocol 1 ($N = 13$), retained in Protocol 2 ($N = 15$). $^{\ddagger}$21.4% retention due to motion; retained under intent-to-treat. Telemetry: NeuSen NSM2 / EEGCap32-pony (1000 Hz, 4-s epochs, CAR, F3, F4, C3, C4, O1, O2; IRB No. 2025-118-01 [1]). Model comparisons across protocols appear in Table S2; operating thresholds are detailed in Note C.

TABLE S2

MATCHED-INPUT MODEL COMPARISONS (AUROC). ALL METHODS SHARE FEATURE COLUMNS, SOURCE-ONLY PARAMETER SELECTION, SCALING, DIRECTION CODING, AND SOURCE-TRAINING MEDIAN THRESHOLDS WITHIN EACH PROTOCOL. TRUNCATION IS A FIXED SCORING-RULE VARIANT ($c = 1.5$). TARGET STANDARDIZATION AND DIRECTION ALIGNMENT ARE UNLABELED BATCH ADAPTATIONS.

| Cohort and protocol | Linear | Clipped | Ridge | Logistic | LDA |
|---|---|---|---|---|---|
| Mendeley, 25 outer folds | 0.9324 | 0.9291 | 0.8813 | 0.8964 | 0.8769 |
| CRRC 15, source frozen | 0.5741 | 0.5741 | 0.5556 | 0.5556 | 0.5556 |
| CRRC 15, target standardized | 0.5000 | 0.5000 | 0.5185 | 0.5185 | 0.5370 |
| CRRC 15, target aligned | 0.7778 | 0.7963 | 0.7407 | 0.7593 | 0.7593 |
| CRRC 13, target aligned | 0.7143 | 0.7143 | 0.7143 | 0.6905 | 0.6905 |
| Figshare 50, $D_1$, source frozen | 0.7691 | 0.7606 | 0.7759 | 0.7674 | 0.7742 |
| Figshare 50, $D_1$, target standardized | 0.7657 | 0.7606 | 0.7708 | 0.7589 | 0.7691 |

TABLE S3

HYPERPARAMETER STABILITY: SOFT TRUNCATION AND T-GECF GATING GRID

**(A) Soft Truncation Threshold $c$**

| Threshold | AUROC | PR-AUC | Bal. Acc. |
|---|---|---|---|
| $c = 1.0\sigma$ | 0.8958 | 0.8543 | 0.8080 |
| $c = 1.5\sigma$ (Default) | **0.9149** | **0.8782** | **0.8399** |
| $c = 2.0\sigma$ | 0.9056 | 0.8662 | 0.8046 |
| $c = 2.5\sigma$ | 0.8996 | 0.8622 | 0.8129 |
| $c = 3.0\sigma$ | 0.8996 | 0.8622 | 0.8129 |
| $c = \infty$ (Raw unconstrained) | 0.8996 | 0.8622 | 0.8129 |

**(B) T-GECF Gating Grid ($\theta_0 \times \kappa$)**

| | Sigmoid Gate Slope $\kappa$ | | | |
|---|---|---|---|---|
| $\theta_0$ (Quantile) | 0.5 | 1.0 | 2.0 | 5.0 |
| 1.282 (90%) | **0.9220** | 0.9182 | 0.9187 | 0.9149 |
| 1.645 (95%) | 0.9160 | 0.9182 | **0.9116** | 0.9089 |
| 1.960 (97.5%) | 0.9160 | 0.9149 | 0.9056 | 0.9056 |
| 2.326 (99%) | 0.9127 | 0.9116 | 0.9022 | 0.8996 |
| Ablate Cont. ($c = \infty$) | 0.8996 | 0.8996 | 0.8996 | 0.8996 |

tracking ensures artifact gaps reset continuity ($\mathbb{I}_{\text{adj}} = 0$); scoring is formally gated until $\geq 10$ adjacent clean pairs accumulate, outputting `INSUFFICIENT_DATA` otherwise. Source-frozen standardized coordinates ($z_k = s_k(x_k - \mu_k)/\sigma_k$) apply archived thresholds ($\theta_0 = 0.0451$ raw, $\theta_{1.5} = 0.0725$ clipped).

Systems engineering requires distinguishing *algorithmic step execution time* from *end-to-end physical pipeline latency*. On pre-buffered periodogram arrays across two complete source records (5,400 epochs each; Table S4), algorithmic execution time per evaluated step was $2.1542 \pm 0.003$ ms (cached power ingest: $0.0098$ ms; feature extraction: $2.1323$ ms; scoring: $0.0110$ ms; max: $3.9261$ ms), confirming bit-level offline agreement ($|\Delta_{\text{feature}}| \leq 2.77 \times 10^{-13}$). Computational operations proceed $> 1{,}800\times$ faster than the 4-s epoch window, introducing zero processor bottlenecks. In live continuous telemetry, physical latency reflects signal processing stages: (1) FIR filter group delay ($\approx 1.95$ s); (2) epoch buffering (4.0 s); and (3) historical adjacency accumulation, requiring a cold-start initialization of $\geq 40$ s ($10 \times$ 4-s clean blocks) and reaching asymptotic estimator stability within $5.0$ minutes ($M^* \approx 75$ blocks). Once initialized, the display updates every 4.0 s with 2.15-ms execution overhead. Complete trajectories are archived in `results/bedside_playback_demo/`.

## REFERENCES


[1] F. Duan, J. Chen, Z. Yan, X. Cao, G. Xiong, and F. Li, "Auditory paradigm-dependent EEG feature screening and representation for distinguishing consciousness states in male patients with disorders of consciousness," *Frontiers in Neurology*, vol. 17, p. 1927313, 2026.


TABLE S4

CACHED-POWER REPLAY TIMINGS AND VERIFICATION (5,400 EPOCHS PER RECORD; TWO RECORDS)

| Processing Phase | Calls/record | Mean (ms) | Mean p95 (ms) | Max (ms) |
|---|---|---|---|---|
| Cached Power Ingest | 5,373 | 0.0098 | 0.0111 | 0.7090 |
| Feature Extraction | 27 | 2.1323 | 3.1117 | 3.9057 |
| Consensus Scoring | 27 | 0.0110 | 0.0123 | 0.0164 |
| Evaluated Step Total | 27 | 2.1542 | 3.1330 | 3.9261 |

Offline Consistency: Max feature $|\Delta| \leq 2.77 \times 10^{-13}$; Score error: 0.000.